First stochastic SPH run: ideal symplectic step + per-particle Metropolis sweep

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Abstract

Working notes for the first stochastic SPH run. One deterministic *ideal* symplectic step is followed by a single Metropolis sweep over the *particles*. Each particle move draws one Gaussian stress ξ_a at particle a, applies it as momentum- and energy-conserving kicks through *all* of a's bonds, and is accepted with $\min(1, e^{\Delta \mathbb{S}_a})$, where $\Delta \mathbb{S}_a = \Delta t\, V_a\, \xi_a : \varepsilon_a$ sees the full SPH strain. *All* dissipation comes from the accept/reject asymmetry; there is no separate deterministic dissipative step. The mean drift of the sweep is *exactly* the deployed two-level viscous force `dissipative_force!`, and the noise covariance is forced by the match — there is no free amplitude and nothing to calibrate. Derivations, accuracy limits, and the implementation plan are collected in the appendices; the end target is the stochastic decay of a 2D sound wave.

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I. SCOPE AND NOTATION

Notation follows `sph2.tex`: particle a carries the extensive entropy s_a and momentum m_a^i; the smoothed entropy field is $S(x_a) = \sum_b s_b W(x_a, x_b)$; the EOS solve yields v_a, γ_a,

$T_{\mathrm{rel},a}$; and the smoothed conjugates give the velocity $\dot{x}_a$, the temperature T_a, and

$$\mathsf{b}_a \equiv \frac{\dot{x}_a}{\mathsf{T}_a} \,. \tag{1}$$

$D_j(x_a, x_b) \equiv \partial W(x_a, x_b)/\partial x_a^j$ is the kernel gradient, with $D_j(x_b, x_a) = -D_j(x_a, x_b)$, and $V_a \equiv s_a/S_a$ is the particle "volume" (the SPH estimate of $\mathrm{d}^d x$, `sph2.tex` App. D).

Three results from `sph2.tex` are used verbatim :

- Density-frame noise covariance (FDT):

$$\left\langle \xi^{ij}(\boldsymbol{x}, t)\, \xi^{lm}(\boldsymbol{x}', t') \right\rangle = 2\, T_{\mathrm{rel}}\, \kappa^{ijlm}(v)\; \delta(t - t')\, \delta^2(\boldsymbol{x} - \boldsymbol{x}') \,. \tag{2}$$

- Tthe entropy change that keeps $\mathbb{H}$ fixed under a momentum kick Δm_a (to linear order; App. C):

$$\Delta s_a = -\,\mathsf{b}_a \cdot \Delta m_a \,. \tag{3}$$

- Total entropy change:

$$\Delta \mathbb{S} = \sum_a \Delta s_a = -\sum_a \mathsf{b}_a \cdot \Delta m_a \,. \tag{4}$$

For deterministic hydro the viscous force was computed in two passes (`sph2.tex` App. D, `dissipative_force!`): Pass 1 builds the SPH strain

$$\varepsilon_{ji}(x_a) \;=\; \frac{1}{S(x_a)} \sum_b s_b \left(\mathsf{b}_{i,b} - \mathsf{b}_{i,a}\right) D_j(x_a, x_b) \tag{5}$$

and Pass 2 assembles the viscous force

$$f^i_{\mathrm{det}}(x_a) = \sum_b s_a s_b\, D_j(x_a, x_b) \left[\left.\frac{T_{\mathrm{rel}}\kappa^{ijlm}}{S^2}\right|_a \varepsilon_{lm}(x_a) + \left.\frac{T_{\mathrm{rel}}\kappa^{ijlm}}{S^2}\right|_b \varepsilon_{lm}(x_b) \right] . \tag{6}$$

We will show how the stochastic step reproduces this mean strain

II. THE STOCHASTIC STEP

A full step takes $(x_a^n, m_a^n, s_a^n) \to (x_a^{n+1}, m_a^{n+1}, s_a^{n+1})$ by an ideal sub-step followed by one per-particle sweep.

a. A — ideal symplectic sub-step. Advance (x_a, m_a) by one step of the *ideal* Hamiltonian flow with the 2nd-order symplectic integrator at fixed Δt (acoustic CFL). No dissipative or stochastic term here: s_a is unchanged and $\mathbb{S}$, $\mathbb{H}$ are conserved to integrator order.

b. B — freeze and strain. Freeze a snapshot of the coefficients $\{S_a, s_a, \mathsf{b}_a, v_a, \gamma_a, T_{\mathrm{rel},a}, \kappa|_a\}$ and precompute the strain ε_a (5) at every particle. During the sweep only the accumulated increments $(\Delta m_a, \Delta s_a)$ change; they are applied to (m_a, s_a) once, at the end of the sweep, and the coefficients are then refreshed for the next step.

c. C — the proposal at particle a. For particle a, draw one symmetric 2×2 Gaussian stress $\xi_a^{ij}=\xi_a^{ji}$, mean zero, with covariance

$$\boxed{\left\langle \xi_a^{ij}\,\xi_a^{lm}\right\rangle \;=\; \frac{2\,T_{\mathrm{rel},a}\,\kappa^{ijlm}(v_a)}{\Delta t\,V_a}\,,\qquad V_a\equiv\frac{s_a}{S_a}} \tag{7}$$

This is the discretised FDT (2) with $\delta(t)\to 1/\Delta t$ and $\delta^d(x)\to 1/V_a$. From ξ_a build a *proposal*: a trial update of a and all of its neighbours, patterned on the particle-a stress piece of the viscous force (6). The proposed momentum changes are, for every bond (a,b) with $r_{ab}<2h$,

$$\boxed{\Delta m_a \mathrel{+}= \Delta t\,\frac{s_a s_b}{S_a^2}\,\xi_a^{ij}D_j(x_a,x_b)\,,\qquad \Delta m_b \mathrel{-}= \Delta t\,\frac{s_a s_b}{S_a^2}\,\xi_a^{ij}D_j(x_a,x_b)\,,} \tag{8}$$

together with $\Delta s_c=-\mathsf{b}_c\cdot\Delta m_c$ (3), for every particle c neighboring a. The proposal conserves momentum bond by bond, conserves $\mathbb{H}$ to linear order in the kick, and has $\Delta m\sim\sqrt{\Delta t}$; positions are untouched. Nothing is applied yet — whether the trial update is kept is decided next.

d. D — accept / reject. The entropy change of the proposed update collapses onto the strain:

$$\begin{aligned}\Delta\mathbb{S}_a &= \sum_c \Delta s_c && (9)\\ &= -\sum_c \mathsf{b}_c\cdot\Delta m_c = \Delta t\,\frac{s_a}{S_a^2}\,\xi_a^{ij}\sum_b s_b\big(\mathsf{b}_{i,b}-\mathsf{b}_{i,a}\big)D_j(x_a,x_b)\\ &= \boxed{\Delta t\;V_a\;\xi_a^{ij}\,\varepsilon_{ji}(x_a)} && (10)\end{aligned}$$

This is the discrete form of $\int \mathrm{d}t\,\mathrm{d}^d x\;\xi^{ij}\partial_j\beta_i$, and of either sign. Note to compute $\Delta\mathbb{S}_a$ at this linearized level we can just the previously stored strain. Accept the proposal with probability

$$\boxed{P_{\mathrm{acc}}(a)=\min\big(1,\;e^{\,\Delta\mathbb{S}_a}\big)\,;} \tag{11}$$

on acceptance the trial $(\Delta m,\Delta s)$ are added to the accumulators; on rejection the proposal is discarded entirely and the state is unchanged. Because $\xi_a\to-\xi_a$ is equiprobable and the covariance depends only on frozen coefficients, this is a plain symmetric-proposal Metropolis move: it preserves the target weight $\propto e^{\mathbb{S}}$ on the $(\mathbb{H},\mathbf{P})$ level set fixed by the ideal flow. Dissipation is entirely the acceptance asymmetry.

e. E — the sweep. Zero the increment accumulators $(\Delta m_a,\Delta s_a)$, then one pass over all particles $a=1\ldots N$, each visited once, accumulating accepted kicks; then fold the accumulators into (m_a,s_a) and refresh. With the freeze the moves are mutually independent, so the sweep is order-free and parallel (App. B). Exactly one sweep per ideal step: the covariance (7) carries the Δt.

III. MEAN DRIFT

Δm and $\Delta\mathbb{S}_a$ are both linear in the Gaussian ξ_a, so the Metropolis average follows from the identity $\big\langle X\min(1,e^Y)\big\rangle=\frac{1}{2}\langle XY\rangle+O(\sigma^3)$ for mean-zero jointly Gaussian (X,Y). Applied

to a single particle move, and summing the sweep — particle a is kicked once by its *own* move (the particle-a stress term) and once, in reaction, by *each neighbour's* move (their particle-b stress terms; the draws are independent, so the means add) — the result is (App. A)

$$\boxed{\left\langle \Delta m_a \right\rangle_{\rm acc} = \Delta t \sum_b s_a s_b \, D_j(x_a, x_b) \left[\frac{T_{\rm rel}\kappa^{ijlm}}{S^2}\bigg|_a \varepsilon_{lm}(x_a) + \frac{T_{\rm rel}\kappa^{ijlm}}{S^2}\bigg|_b \varepsilon_{lm}(x_b) \right] = \Delta t \, f^i_{\rm det}(x_a) \,,} \tag{12}$$

the **full** two-level force (6). Two cancellations in the derivation are structural: the Metropolis $\frac{1}{2}$ (only the downhill half of the noise is suppressed) exactly eats the FDT factor of 2 in (7), and the $1/V_a$ from the discretised $\delta^d(x)$ supplies the volume power that a naive per-bond force misses (App. D). The covariance (7) is therefore *forced* by the match: there is no free amplitude, and the shear-wave and sound-wave runs are *validations*, not calibrations.

IV. DIAGNOSTICS — WHAT "WORKING" MEANS

- **Momentum:** $\sum_a m^i_a$ constant to machine precision.
- **Energy:** $\mathbb{H}$ shows the *expected* small positive drift at relative rate $\sim 1/(s_a \tau_{\rm visc})$ [App. C] on top of the integrator error — linear in t, with the predicted magnitude; anything faster is a bug.
- **Second law:** $\langle \Delta \mathbb{S} \rangle \geq 0$ per sweep; individual moves go both ways.
- **Acceptance:** near unity ($\gtrsim 0.99$) in the small-kick regime; it is *not* tuned. Low acceptance means Δt is too large for the given η/s and gradients.
- **Grid-scale modes:** rms(m_a) on a quiescent uniform background must *saturate* at the FDT equipartition value ($\langle v_a^2 \rangle \sim 1/s_a$; requires $s_a \gtrsim 10^3$ for non-relativistic thermal fluctuations), in contrast to the per-bond $\sqrt{t}$ runaway (App. D).
- **Equilibrium fluctuations:** time-averaged $\langle \delta S^2 \rangle$, $\langle m^2 \rangle$ vs. the FD prediction; repeat at $\Delta t/2$ to bound the $O(\Delta t)$ weak bias [App. C].
- **Mean drift:** sweep-averaged $\Delta m_a/\Delta t$ on a frozen sheared state matches `dissipative_force!` — (12) head-to-head.
- **Shear wave (validation):** decay rate of $M^y = A\sin(kx)$ matches $\gamma_k = (\eta/s)\, k^2/T_0$ with no tuned parameter.
- $\eta/s \to 0$**:** $\xi_a \to 0$; the run reduces to the deterministic ideal symplectic integrator.

V. FIRST RESULTS ON A FROZEN LATTICE

The unit checks of `teststochastic.jl` all pass: momentum is conserved to round-off, the sweep is visit-order independent, and the mean drift of the sweep reproduces `dissipative_force!` (slope 0.994 ± 0.007). The uniform-background run `examples/teststochastic_run.`

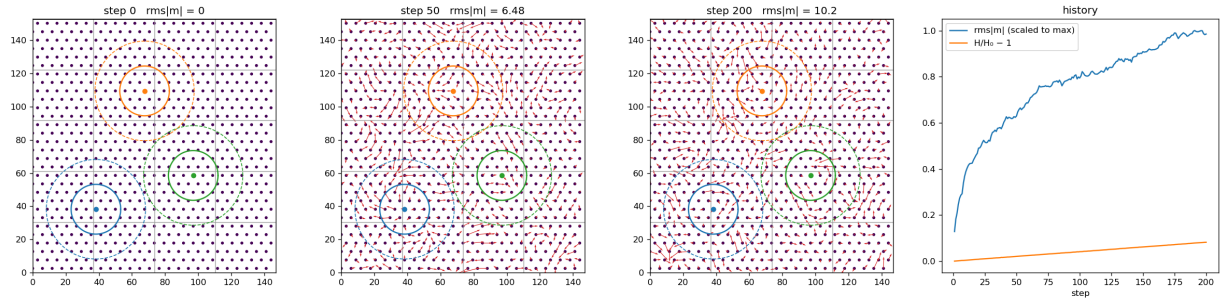


FIG. 1. Cold start, frozen lattice (`simple_cold_start.jl`). Panels 1–3: particles at steps 0, 50, 200; arrows are m_a (rescaled per panel, so compare the rms values in the titles, not arrow lengths). Dots are coloured by $S(x_a)$, uniform as it should be. Grey lines: cell-list cells. Circles of radius h (solid) and $2h$ (dashed, the kernel support) around three particles. Panel 4: rms$|m|$ (scaled to its maximum) and $\mathbb{H}/\mathbb{H}_0 - 1$ vs step.

however, fails. Started at the FDT equilibrium with the ideal step on, after 3000 steps rms(M) is 157 times the FDT value and $\mathbb{H}$ has grown by 115%. (Before the cell-list fix described below it blew up at step 419.) This section isolates why, with the particles held fixed so that only the stochastic sweep acts.

a. Setup (all figures). 2D conformal fluid, $c_s^2 = 1/2$, units $\ell_Q = T_0 = 1$, $s_0 = 1$, $\eta/s = 10/\sqrt{2} \approx 7.07$, $\zeta = 0$. Hex lattice with $h = 15$, $h/a = 2.55$ ($a = 5.88$), box 147.1×152.8, $N = 750$, particle volume $V = 29.97$, $s_a \approx 30$, $T_{\rm rel} = 0.912$, $w = e + p = 0.759$; each particle has 91 neighbours within $2h$. $\Delta t = 4.24$ (acoustic CFL 0.2) unless stated. *Frozen lattice* means no ideal step: positions never move, and each step is refresh fields → `particle_sweep!` → `apply_sweep!`. *Cold start* means all $m_a = 0$ initially: all of the energy is internal, none kinetic. The scripts are in `examples/` and named in each caption.

b. A cell-list bug, now fixed. Drawing the cells (Fig. 1) exposed a bug in `CellList2D`: it used one square cell width L_x/n_x in both directions. In this box $L_y/n_y < L_x/n_x$, so the rows of cells did not tile $[0, L_y)$, the periodic y-stencil did not connect the top and bottom rows, and 5% of the neighbour pairs (all across $y = 0$) were dropped; $S(x_a)$ fell to 0.83 there on a uniform lattice. The cells now have separate widths $(L_x/n_x,\ L_y/n_y)$; `test/testcelllist_box.jl` checks this. All results below are with the fix. The fix changes the numbers by $\sim 20\%$ but neither of the two problems below.

A. Cold start: what the sweep does to particles at rest

Fig. 1 is the most basic run. The noise immediately builds a momentum field that changes direction from one particle to the next, on the lattice spacing a, which is much shorter than h. Inside a single h-circle (about 20 particles) the arrows point every which way, so the smoothed field $M(x) = \sum_b m_b W(x - x_b)$ averages most of this momentum away. rms$|m|$ is 6.5 at step 50 and 10.2 at step 200 and is still rising. A working scheme would saturate at the FDT value. $\mathbb{H}$ rises in a straight line (+8.2% by step 200). With $\mathbb{H}$ fixed, the fluctuation kinetic energy should instead come *out* of the internal energy: by equipartition about T per particle against $s_a T \approx 30T$ internal, a few percent of $\mathbb{H}$.

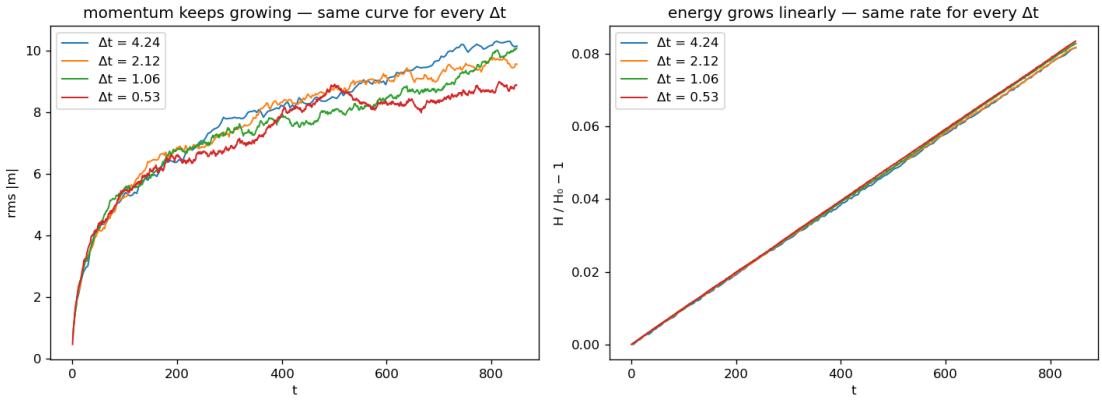


FIG. 2. Cold start, frozen lattice, four time steps run to the same physical time $t \approx 848$ (`dt_scan.jl`). Left: rms$|m|$. Right: $\mathbb{H}/\mathbb{H}_0 - 1$.

B. The failures do not go away as $\Delta t \to 0$

Fig. 2 repeats the cold start at $\Delta t = 4.24, 2.12, 1.06, 0.53$, always to the same physical time. The curves lie on top of each other (at $t \approx 848$: rms$|m| = 10.2, 9.6, 10.1, 8.9$ and $\mathbb{H}/\mathbb{H}_0 - 1 = 0.082, 0.082, 0.083, 0.084$), and the acceptance goes to 1 as $\Delta t \to 0$ ($0.961 \to 0.986$). The scheme is converging to a well-defined continuous-time limit; it is that limit that is wrong. The $O(\Delta t)$ weak biases of App. C(a) would shrink, so these are not them. Two separate defects:

1. **Energy.** $\mathbb{H}$ grows at $\approx 1.0 \times 10^{-4}$ per unit time, independent of Δt. This is the linearized $\Delta s = -\mathsf{b} \cdot \Delta m$ of App. C(b). The dropped term $\frac{1}{2}\Delta m \cdot \partial^2 \mathbb{H} \cdot \Delta m$ is $O(\Delta t)$ per step and always positive, so it adds up to an $O(1)$ drift in physical time. It is suppressed by $1/s_a$, and $s_a = 30$ here is not large.

2. **Momentum.** rms$|m|$ keeps growing in the continuous-time limit. The rest of this section shows why.

C. What the smoothing sees: the response $R(k)$

Everything in the scheme ($\mathbb{H}$, $\dot{x}$, b, the strain, the accept test) sees the particle momenta only through $M(x_a)$. Put a wave on the particles, $m_b = \hat{m}\cos(\mathbf{k} \cdot \mathbf{x}_b)$. Converted back to a momentum per particle, the smoothed field returns it scaled:

$$V\, M(x_a) = R(\mathbf{k})\, \hat{m} \cos(\mathbf{k} \cdot \mathbf{x}_a)\,, \qquad R(\mathbf{k}) = V \sum_b W(x_a - x_b) \cos\big(\mathbf{k} \cdot (\mathbf{x}_b - \mathbf{x}_a)\big)\,. \tag{13}$$

$R(\mathbf{k})$ is the fraction of the wave the smoothed field keeps. Fig. 3 evaluates (13) on the actual neighbours. $R(0) = 1.0002$ (a uniform flow is fully seen), and the lattice sum matches the continuum $\tilde{W}(k)$ up to $|k| \approx 1$. R falls to 0.5 at $|k| \approx 0.15$, to 0.1 at $|k| \approx 0.26$, and is essentially zero beyond $|k| \approx 0.35$ (a wavelength of about $1.2h$). On a lattice, $\mathbf{k}$ and $\mathbf{k} + \mathbf{G}$ give the same pattern, which is why R returns to 1 at the reciprocal vectors

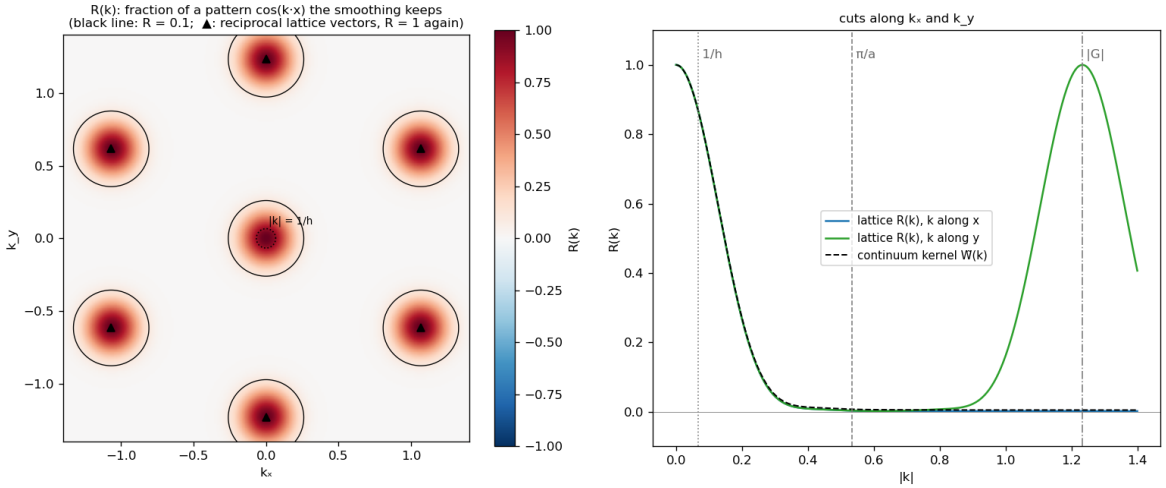


FIG. 3. Smoothing response of the lattice (`smoothing_response.jl`). Left: $R(\mathbf{k})$ in the plane; black contour $R = 0.1$; dotted circle $|k| = 1/h$; triangles: reciprocal lattice vectors (copies of $k = 0$). Right: cuts along k_x and k_y compared with the continuum kernel transform $\tilde{W}(k)$.

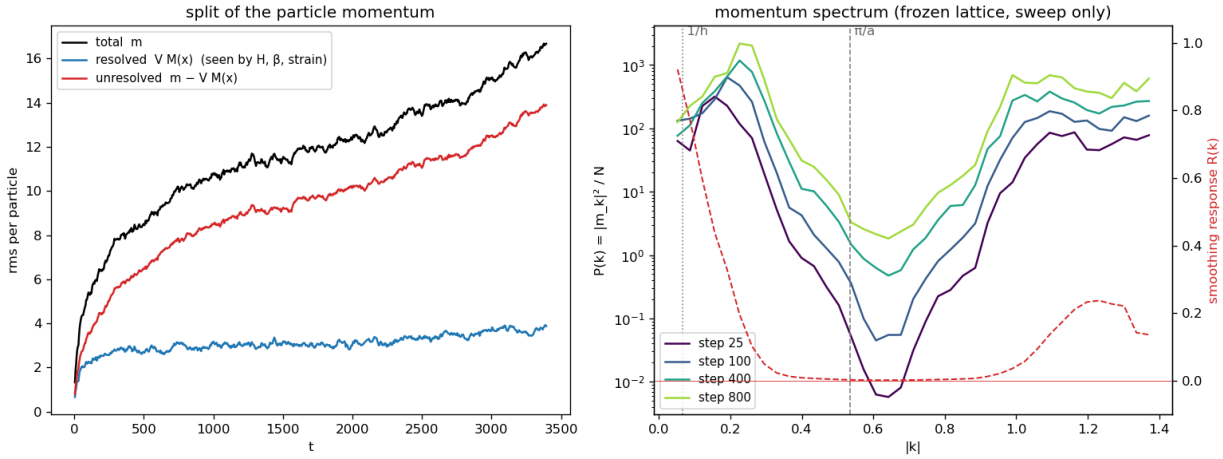


FIG. 4. Cold start, frozen lattice, 800 steps (`momentum_split.jl`). Left: rms of the total momentum, its resolved part $V_a M(x_a)$ and its unresolved part $m_a - V_a M(x_a)$. Right: spectrum $P(\mathbf{k}) = |\sum_a m_a e^{-i\mathbf{k}\cdot\mathbf{x}_a}|^2/N$ binned in $|k|$ at four times, with R (red dashed, right axis) averaged in the same bins. The bump at $|k| > 1$ is lattice aliasing of the low-k modes (Fig. 3).

($|G| = 4\pi/\sqrt{3}a = 1.23$). Every distinct pattern has a $\mathbf{k}$ in one hexagonal zone of area ≈ 1.32, and $R > 0.1$ only in a disk of area ≈ 0.21. So **only about** 16% **of the lattice's momentum patterns are seen by the smoothing at more than** 10%. This is the price of $h/a = 2.55$: each smoothed value averages over about 90 neighbours.

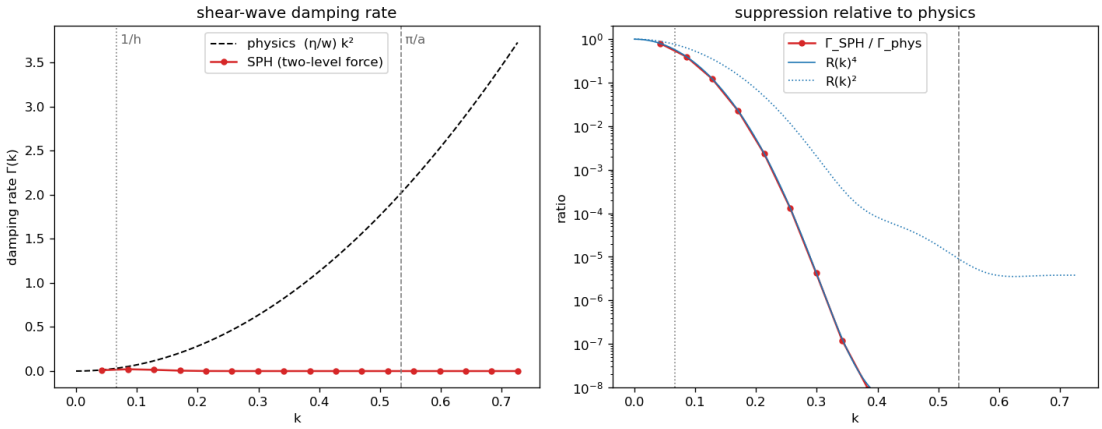


FIG. 5. Deterministic damping rate of a shear wave on the frozen lattice (`damping_rate.jl`). Left: $\Gamma(k)$ for the two-level force and the physical $(\eta/w)k^2$. Right: the ratio, with $R(k)^4$ and $R(k)^2$.

D. The growing momentum is the part the smoothing does not see

Split each particle's momentum into the part the smoothed field sees and the rest:

$$m_a = \underbrace{V_a M(x_a)}_{\text{resolved}} + \underbrace{\left[m_a - V_a M(x_a)\right]}_{\text{unresolved}}, \qquad V_a = s_a/S_a\,. \tag{14}$$

For a long wave the resolved part is all of it; for a pattern that changes from particle to particle it is nearly none. Fig. 4 (left) shows that the resolved part levels off at about 3 and barely moves. All of the growth is unresolved: at step 800 the rms values are total 16.7, resolved 3.9, unresolved 13.9. The spectrum (right) shows where the power piles up: $|k| \approx 0.15$–0.3, wavelengths around $2h$, where R is small but not zero. Near the zone boundary, where $R \approx 0$, there is little power, because the kicks $\xi_a D(x_a, x_b)$ are built from the kernel too and barely excite those patterns.

E. The damping rate: $\Gamma_{\rm SPH} = \Gamma_{\rm phys}\, R^4$

This measurement has no noise and no time stepping, and it does not start from rest. On the frozen lattice put a small shear wave in the momenta, $m_a = A\,\hat{y}\cos(kx_a)$ with A chosen so $v \sim 10^{-4}$. Refresh the fields, evaluate the deterministic two-level force `dissipative_force!` once, and project it onto the wave:

$$\Gamma_{\rm SPH}(k) = -\frac{\sum_a f_a \cdot m_a}{\sum_a m_a \cdot m_a}\,. \tag{15}$$

Since the mean drift of the sweep is this force (Sect. III), this is also the damping rate of the stochastic scheme. Physically a shear wave decays at $\Gamma_{\rm phys} = (\eta/w)k^2$, so the shortest waves die fastest. Fig. 5 and Table I show that the scheme does the opposite:

$$\boxed{\Gamma_{\rm SPH}(k) = \Gamma_{\rm phys}(k)\, R(k)^4} \tag{16}$$

k	$2\pi/(kh)$	$R(k)$	$\Gamma_{\mathrm{SPH}}/\Gamma_{\mathrm{phys}}$	$R(k)^4$
0.043	9.8	0.945	0.795	0.796
0.085	4.9	0.795	0.397	0.398
0.171	2.5	0.388	2.2×10^{-2}	2.3×10^{-2}
0.256	1.6	0.108	1.3×10^{-4}	1.3×10^{-4}
0.342	1.2	0.019	1.2×10^{-7}	1.3×10^{-7}

TABLE I. Shear-wave damping on the frozen lattice ($2\pi/(kh)$ is the wavelength in units of h).

to three digits for every $k \lesssim 0.34$ (beyond that both are below 10^{-7}). The rate peaks near $k \approx 0.1$ and then goes to zero. The four powers of R have two origins:

- R^2 **from the representation**: $\mathsf{b} = \dot{x}/\mathsf{T}$ is built from m by two smoothings ($M = \sum mW$, then $\dot{x}_a = \sum_b s_b (v/S)_b W$), so $\mathsf{b}_{\mathbf{k}} \propto R^2 m_{\mathbf{k}}$.
- R^2 **from the discretization of the dissipative function**: the strain (5) is a kernel gradient of b ($\propto kR$), and the force (6) is a second kernel divergence ($\propto kR$). This "gradient of a smoothed gradient" Laplacian is a wide stencil, the SPH version of odd–even decoupling. A Laplacian built from two centred first differences also fails to damp the checkerboard mode.

a. Consequences.

- **Deterministic runs are affected too.** A shear wave with wavelength $5h$ decays at 40% of the physical rate, and one with $10h$ at 80%. The deterministic shear- and sound-wave validations should be rechecked against this before anything is calibrated.
- **Why the stochastic momentum grows.** The following scalings are *derived, not measured*; only the rate is measured. For a mode $\mathbf{k}$ the noise injected per unit time goes as k^2R^2 (the kick pattern), the friction rate as k^2R^4 (16), and the equilibrium amplitude as $\langle |m_{\mathbf{k}}|^2 \rangle \propto 1/R^2$ (its energy cost goes through $M = \sum mW$). For unresolved modes the target is huge and the approach to it is very slow, so they simply keep growing. Note that K sets only the rates: the equilibrium is fixed by $\mathbb{H}$ and $\mathbb{S}$.
- **Relation to App. D.** The per-bond scheme failed through the same mechanism. Its friction acted through the smoothed b, so it was suppressed by R^2, while its noise was not suppressed, so the runaway was fast. The two-level move puts in less noise (k^2R^2), which slows the runaway, but it damps by an additional R^2. It is not a remedy. In App. D the claim that "the strain-based friction acts on what is excited" is false.

F. A one-level (bond) form

The two-level K damps unresolved modes as R^4, and has an exact null mode at the zigzag. A one-level (bond) form squares the differences of b across neighbouring pairs instead, after subtracting a mean-field rotation and dilation from each. It has the same continuum limit, including the full $\kappa^{ijlm}(v)$ of a moving fluid, conserves momentum and angular momentum, and has no null mode. It is derived in `bond_dissipation.tex` and is selected with

`params.dissipation = :bond`. On the smoothed `b` its damping still levels off at the floor $\sim R^2$, which comes from `b` and not from K. The default, `:changerep`, removes this floor by doing the viscous step in a particle representation (`changerep.tex`). The matching noise is implemented only for the two-level form, which the stochastic sweep therefore uses.

No choice of K alone can fix this, because K is a quadratic form in `b` and `b` carries R^2. The particle-level momentum needs its own thermodynamic force. That means changing $\mathbb{H}$ or $\mathbb{S}$: for example the energy representation with particle entropy $\sum_\ell S(e_\ell, m_\ell)$, for which concavity gives $S(\underline{P}(x)) \geq \underline{S}(x)$ (`sph1.tex`, "Convexity").

Appendix A: Derivation of the mean drift

Write the kick to a from its own move as $X = \Delta m_a^i$ and the accept exponent as $Y = \Delta\mathbb{S}_a$; both are linear in ξ_a with frozen coefficients, so (X, Y) are jointly Gaussian and mean zero, and

$$\big\langle X\, \min(1, e^Y)\big\rangle = \tfrac{1}{2}\,\langle XY\rangle + O(\sigma^3) \tag{A1}$$

[split $\min(1, e^Y) = 1 + (e^Y - 1)\mathbb{1}[Y < 0]$ and use $\langle XY\,\mathbb{1}[Y<0]\rangle = \frac{1}{2}\langle XY\rangle$ for a mean-zero Gaussian pair; the remainder is $O(\sigma_Y^3) = O(\Delta t^{3/2})$]. With $G_j \equiv \sum_b s_b D_j(x_a, x_b)$ this gives

$$\big\langle \Delta m_a^i \big\rangle_{\rm acc} = \tfrac{1}{2}\,\Delta t^2\,\frac{s_a^2}{S_a^3}\,G_j\,\varepsilon_{lk}(x_a)\,\big\langle \xi_a^{ij}\xi_a^{kl}\big\rangle + O(\xi^3)\,. \tag{A2}$$

Substitute the covariance (7) with index pairs (ij), (kl),

$$\big\langle \xi_a^{ij}\xi_a^{kl}\big\rangle = \frac{2\,T_{{\rm rel},a}\,\kappa^{ijkl}\big|_a}{\Delta t\,V_a}\,, \qquad V_a = \frac{s_a}{S_a}\,, \tag{A3}$$

and collect the prefactors:

$$\begin{aligned}\big\langle \Delta m_a^i \big\rangle_{\rm acc} &= \underbrace{\tfrac{1}{2}\cdot 2}_{=1}\,\underbrace{\frac{\Delta t^2}{\Delta t}}_{=\Delta t}\,\underbrace{\frac{s_a^2}{S_a^3}\cdot\frac{S_a}{s_a}}_{=s_a/S_a^2}\,T_{{\rm rel},a}\,\kappa^{ijkl}\big|_a\,\varepsilon_{lk}(x_a)\,G_j \\ &= \Delta t\,\frac{s_a}{S_a^2}\,T_{{\rm rel},a}\,\kappa^{ijkl}\big|_a\,\varepsilon_{lk}(x_a)\,G_j\,.\end{aligned} \tag{A4}$$

The Metropolis $\frac{1}{2}$ cancels the FDT 2, and the $1/V_a$ removes one power of the particle volume.

Next the index contraction. In (A4) the strain carries the Pass-1 convention ε_{lk} = (gradient index first, velocity index second), so κ's *last* index l meets the gradient slot and its *third* index k the velocity slot — the opposite order from the deterministic contraction $\kappa^{ijlm}\varepsilon_{lm}$. One dummy relabel ($k \to m$) and the symmetry of κ in its last pair ($\kappa^{ijml} = \kappa^{ijlm}$, true of the $2\eta(\cdots) + \zeta(\cdots)$ structure) fix this:

$$\kappa^{ijkl}\,\varepsilon_{lk} \overset{k\to m}{=} \kappa^{ijml}\,\varepsilon_{lm} \;=\; \kappa^{ijlm}\,\varepsilon_{lm}\,; \tag{A5}$$

in particular only the *symmetric* part of ε survives, as in the deterministic force. With G_j restored,

$$\big\langle \Delta m_a^i \big\rangle_{\rm acc} = \Delta t\,\frac{s_a}{S_a^2}\,\big(T_{\rm rel}\kappa^{ijlm}\big|_a\,\varepsilon_{lm}(x_a)\big)\,\sum_b s_b\,D_j(x_a, x_b)\,, \tag{A6}$$

precisely the particle-a stress contribution to the two-level force (6).

This is half of the story: (A6) is the mean kick a receives *from its own move*. But a is also a neighbour of each b, so it is kicked again — in reaction — every time b does *its* move. Bond (b, a) is one of b's bonds, so b's move applies

$$\Delta m_a \mathrel{-}= \Delta t \, \frac{s_a s_b}{S_b^2} \, \xi_b^{ij} \, D_j(x_b, x_a) \;=\; +\Delta t \, \frac{s_a s_b}{S_b^2} \, \xi_b^{ij} \, D_j(x_a, x_b) \tag{A7}$$

(kernel antisymmetry), and this kick is accepted with b*'s* exponent $\Delta \mathbb{S}_b = \Delta t \, V_b \, \xi_b^{kl} \varepsilon_{lk}(x_b)$ — the same ξ_b in both, so (A2)–(A6) go through with every coefficient evaluated at b:

$$\left\langle \Delta m_a^i \right\rangle_{\text{acc}}^{(b\text{'s move})} = \Delta t \; s_a s_b \, D_j(x_a, x_b) \, \frac{T_{\text{rel}} \kappa^{ijlm}\big|_b \, \varepsilon_{lm}(x_b)}{S_b^2} \, . \tag{A8}$$

The draws ξ_a, $\{\xi_b\}$ are independent, so over one full sweep the means add — a is touched once as owner and once per neighbour as reactor — giving (12).

Appendix B: Sweep order, parallelism, and exactness

With the coefficient freeze and the accumulator fields — no in-place update of m, s — each particle's proposal ξ_a, its kicks, and its accept test depend only on frozen coefficients, the precomputed ε_a, and that particle's own draws (ξ_a, U_a). Every particle carries its **own Xoshiro stream** `rngs`$[a]$, a vector of N generators seeded once per run from a master `Xoshiro(seed)` (`seed` from `SimParams`). The stream is tied to the particle index, not to when the particle is visited, and the streams advance from one sweep to the next. Nothing one move reads is written by another; the sole inter-move coupling is the additive accumulation into $(\Delta m, \Delta s)$, which commutes.

a. Order-independence. Because particle a always draws from `rngs`$[a]$, a single pass yields the same $(\Delta m, \Delta s)$ regardless of visit order — fixed, random-permuted, or fully parallel. The accepted set and every kick are *bit-identical* across orderings; the per-particle sums differ only by floating-point associativity ($\sim$ a few ulps).

b. One move per particle. Every particle is visited exactly once per sweep, so the noise/dissipation dose per Δt is spatially uniform by construction. Non-uniform density needs no attempt-weight correction: the covariance (7) carries $1/V_a$ and the strain (5) carries $1/S_a$, so the density dependence is exactly that of the deterministic force.

c. Synchronous (parallel) sweep. Decoupled decisions + additive writes: propose and decide all particles in parallel; reduce the accepted kicks (atomics or per-thread partials); apply. The per-particle stream vector makes every kick and accept flag independent of thread/rank count (particle a's noise is fixed no matter which thread runs its move); only the reduction order (the last few ulps) changes. Distributed memory: standard MD halo — exchange positions and frozen coefficients within $2h$ of each subdomain boundary (the strain pass needs b within $2h$; the kicks reach $2h$), compute owned particles, reduce boundary contributions to owners.

d. Exactness: Metropolis-within-Gibbs. The frozen synchronous sweep is *not* an exact sampler of $e^{\mathbb{S}}$: each particle's accept test ignores the kicks applied by the other moves in the same sweep (their stencils overlap within $2h$), so the composite kernel preserves $e^{\mathbb{S}}$ only up

to the standard $O(\Delta t)$ weak (Euler–Maruyama-type) error. The classical exact alternative is *sequential Metropolis-within-Gibbs*: visit particles in order, fold each accepted move into the live state and refresh b_c of the touched particles before the next move — each single-particle kernel then satisfies detailed balance and the composite preserves the target. But even this is exact only if the state dependence of the proposal covariance (7) ($T_{\mathrm{rel},a}$, $\kappa(v_a)$, V_a all move with the state) is included through the Gaussian Hastings ratio; dropping it is again an $O(\Delta t)$ bias (the usual multiplicative-noise ambiguity). Since *both* variants carry $O(\Delta t)$ weak bias, the frozen sweep — fully parallel and order-independent — loses nothing in accuracy order; the sequential + Hastings variant is the recorded upgrade path if precision equilibrium statistics are ever needed.

Appendix C: Known biases and their sizes

None of the following affects the sound-wave decay target at leading order; they are recorded so the diagnostics test for the *expected* deviations.

a. (a) Stationary distribution: $O(\Delta t)$ weak bias. From the coefficient freeze (App. B). Affects equilibrium fluctuation amplitudes at $O(\Delta t)$, not the mean drift. Detect by running the equilibrium test at Δt and $\Delta t/2$; cure, if ever needed, by sequential Metropolis-within-Gibbs with the Hastings ratio.

b. (b) Energy bookkeeping: the linearized Δs heats at $O(1/s_a)$. eq. (3) is the *first-order* term of "choose Δs so that $\mathbb{H}$ is conserved":

$$\Delta\mathbb{H} = \underbrace{\dot{x}_c \cdot \Delta m_c + \mathsf{T}_c\, \Delta s_c}_{=0} + \tfrac{1}{2}\, \Delta m \cdot \frac{\partial^2 \mathbb{H}}{\partial m^2} \cdot \Delta m + \dots \tag{C1}$$

Deterministically $\Delta m \sim \Delta t$ and the dropped term is a harmless $O(\Delta t^2)$ — which is why `dissipative_step!` gets away with the same formula. Stochastically $\Delta m \sim \sqrt{\Delta t}$, so the dropped term is $O(\Delta t)$ per step and *sign-definite* (positive kinetic Hessian): every landed kick heats by $\sim \Delta m^2/2m_{\mathrm{eff}}$, which summed is the FDT noise-injection power. The friction work is booked into s correctly; the noise heating is booked nowhere, so $\mathbb{H}$ drifts monotonically upward at the relative rate

$$\frac{\dot{\mathbb{H}}}{\mathbb{H}} \;\sim\; \frac{1}{s_a\, \tau_{\mathrm{visc}}}\,, \tag{C2}$$

suppressed by the per-particle entropy $s_a \gtrsim 10^3$. Treat $\Delta\mathbb{H}(t)$ as an *expected-drift diagnostic*: small, positive, linear in t, with this rate — its magnitude is itself a check of the noise normalization. Upgrade if ever needed: midpoint evaluation $\Delta s_c = -\frac{1}{2}[\mathsf{b}_c(m) + \mathsf{b}_c(m + \Delta m)] \cdot \Delta m_c$ (residual $O(\Delta m^3)$), at the cost of a velocity re-solve per accepted move. The mean drift (12) is unaffected: a quadratic addition to $\Delta\mathbb{S}_a$ shifts the Metropolis mean only through vanishing Gaussian third moments.

c. (c) The realized increments are non-Gaussian — by design. The accepted kick distribution is the Gaussian proposal *tilted* by the acceptance: in the scalar toy ($\Delta\mathbb{S} = c\xi$, $\xi \sim N(0, \sigma^2)$) it is the untouched Gaussian on the favorable side, a Gaussian *shifted* to $c\sigma^2$ on the unfavorable side ($N(\xi)e^{c\xi} \propto e^{-(\xi - c\sigma^2)^2/2\sigma^2}$), plus a rejection point mass at zero. Its mean $\frac{1}{2}c\sigma^2$ *is* the drift — the tilt cannot be removed without removing the dissipation.

Higher cumulants are $O(\Delta t^{3/2})$ and vanish in the diffusion limit. Additionally, the linearized $\Delta\mathbb{S}_a$ mis-weights rare large kicks by $e^{O(k^2\Delta t/\tau)}$ at k standard deviations, distorting only far tails ($k \sim \sqrt{\tau/\Delta t}$) that are never sampled. Only mean and covariance survive at $O(\Delta t)$ — both are exactly right by Sect. III.

Appendix D: Why not per-bond accept/reject (record)

The first design used one Metropolis test per *bond*: a pair kick $\delta_{ab} \propto \Delta t\, \mathcal{A}\sqrt{s_a s_b}\,(\xi_a/S_a + \xi_b/S_b)D(a,b)$ accepted with $\min(1, e^{-(\mathsf{b}_a-\mathsf{b}_b)\cdot\delta_{ab}})$. It was coded, tested, and deleted — it fails structurally; do not revisit:

- **The mean is the single-level force, always.** The accept probability sees only δ_{ab}, so $\langle\delta_{ab}\rangle_{\text{acc}} \propto \langle\delta_{ab}\delta_{ab}\rangle\ \Delta\mathsf{b}^{(ab)}$ is a single-bond object with *coincident* kernel gradients $D_j(a,b)D_l(a,b)$ — never the two-level strain. No prefactor choice changes this.
- **The single-level force is dimensionally** $1/\text{vol}$ **too big:** $D_l(a,b)\,\Delta\mathsf{b}^{(ab)}$ is not a gradient estimator ($\sim [T^2]$ where $\partial_l\mathsf{b}$ is dimensionless); the amplitude $\mathcal{A}$ had to absorb a volume factor and be calibrated.
- **Odd-even runaway.** On a quiescent uniform background the per-bond kicks inject independent grid-scale momentum that the *smoothed* b_a cannot see, so the accept/reject provides no friction on those modes: $\text{rms}(m_a) \propto \sqrt{t}$ without bound at every amplitude (observed), while the resolved (smoothed) sector stayed flat.

The per-particle move fixes all three at once: the accept test sees the full strain ε_a [(10)], the mean is the two-level force with the volume power correct (App. A), and the injected pattern $\xi_a\, D(x_a, x_b)$ is smooth in b (a kernel-gradient dipole), so the grid-scale modes are excited far less and the strain-based friction acts on what is excited.

a. Correction (see Sect. V). The last claim is false. The friction on a mode $\mathbf{k}$ is $\Gamma_{\text{phys}}R(k)^4$ (measured, Fig. 5). Two powers of R come from the smoothed b, which the per-bond scheme shared, and two come from the wide two-level stencil. The per-particle move suppresses the noise in those modes, which slows the runaway, but suppresses the friction even more. The per-particle sweep still shows the per-bond symptom (resolved sector flat, unresolved momentum growing), only more slowly.

Appendix E: Implementation

Reused unchanged: `sph.jl` (`Particle`, EOS, `kappa_tensor`), `celllist.jl` (`leapfrog_step!`, cell-list machinery, `dissipative_force!` Pass 1, `wendland_c2_grad`, `minimum_image`), `setup.jl` (IC helpers, `thermalize!`), `io.jl` (`SnapshotWriter`). New code lives in `stochastic.jl` + driver scripts.

a. Noise draw in code. `generate_noise` (`sph2.tex` App. E1) returns the unit-normalized stress $\hat{\xi}$ with $\left\langle \hat{\xi}^{ij}\hat{\xi}^{lm}\right\rangle = \kappa^{ijlm}(v)$ (*no* T_{rel}); `draw_xi` rescales by $\sqrt{2T_{\text{rel}}}$ to the continuum FDT normalization 2`kappa_tensor` (verified by MC audit, `teststochastic.jl` Step 2, at all v). The per-particle draw of (7) is then `draw_xi`$/\sqrt{\Delta t\, V_a}$.

b. The full loop.

1. IC: uniform background, small periodic box (`setup_ic` / `setup_shear_wave`); `thermalize!`.

2. Per step: `leapfrog_step!` → `rebuild_cell_list!` → refresh+freeze (`smooth_to_particles!`, `update_thermodynamics!`, `update_velocities!`) → strain pass → `particle_sweep!` → `apply_sweep!` → diagnostics.

c. Status. Done: accumulator fields + `apply_sweep!` (Step 0); the cell-list neighbour stencil — the traversal used by the strain pass and the sweep — validated against $O(N^2)$ brute force (Step 1; requires $\texttt{ncx}, \texttt{ncy} \geq 3$); `draw_xi` + MC covariance audit (Step 2). The per-bond sweep and its bond list were coded, tested, and deleted (App. D).

```
particle_sweep!(particles, params, cl, eos, dt, rngs=PHYS_RNG) -> (n_acc, n_try, ΔS_tot)
  # coefficients frozen; ε_a from the strain pass (dissipative_force! Pass 1)
  # rngs[a] : particle a's own Xoshiro (rng.jl), seeded once per run
  for a in eachindex(particles); particles[a].Δm = 0; particles[a].Δs = 0; end
  for a in eachindex(particles)
      rng = rngs[a]
      Va  = particles[a].s / particles[a].S_smooth
      ξa  = draw_xi(rng, particles[a], eos, cs2v) / sqrt(dt * Va)   # cov: eq (persite_cov)
      ΔS  = dt * Va * sum(ξa[i,j]*ε_a[j,i] for i,j)          # = dt V_a ξ:ε   eq (dSa)
      n_try += 1
      if rand(rng) < exp(ΔS)                                  # test BEFORE bond loop
          for b in neighbors(a), r_ab < 2h                   # cell-list stencil
              grad = wendland_c2_grad(x_a - x_b, ...)
              f    = dt * (s_a*s_b / S_a^2) * (ξa * grad)  # SVector{2}     eq (kicks)
              particles[a].Δm += f ; particles[a].Δs -= dot(b_vec[a], f)
              particles[b].Δm -= f ; particles[b].Δs += dot(b_vec[b], f)
          end
          n_acc += 1 ; ΔS_tot += ΔS
      end
  end
```

The accumulators are zeroed at entry (a full pass first — a particle is written as a neighbour of others' moves), so no caller-side reset is needed; `apply_sweep!` folds $(\Delta m, \Delta s)$ into (m, s) afterwards. The accept test needs only ξ_a and ε_a [(10)] — no neighbour loop — so rejected proposals cost $O(1)$. *Checks* (extend `teststochastic.jl`): (i) $\eta/s = 0 \Rightarrow$ all $\Delta m, \Delta s = 0$; (ii) $|\sum_a \Delta m_a| / \sum_a |\Delta m_a| < 10^{-12}$; (iii) GENERIC degeneracy $\sum_a (\mathsf{T}_a \Delta s_a + \dot{x}_a \cdot \Delta m_a) = 0$ to round-off; (iv) per-particle stream vector $\Rightarrow$ visit-order independence (shuffled particle order reproduces $(\Delta m, \Delta s)$ to $< 10^{-12}$); (v) the identity (10): the cheap $\Delta t V_a \, \xi : \varepsilon$ equals the explicit $-\sum_c \mathsf{b}_c \cdot \Delta m_c$ over the distributed kicks, per particle, to round-off; (vi) **mean drift:** many sweeps on one frozen sheared state, $\langle \Delta m_a \rangle / \Delta t$ vs. `dissipative_force!` — slope 1, the decisive test of (12).

e. Then. Uniform-background stability (`teststochastic_run.jl` rewired): $s_a \gtrsim 10^3$; rms(m_a) saturates at the FDT value; $\mathbb{H}$ drift positive at the App. C(b) rate; acceptance $\gtrsim 0.99$; repeat at $\Delta t/2$. *Shear-wave validation* (`teststochastic_shear.jl`): ensemble-averaged decay of $M^y = A\sin(kx)$ vs. $(\eta/s)k^2/T_0$, no tuned parameter; an $\eta/s = 0$ run measures the numerical baseline. *2D sound-wave decay — the target* (`teststochastic_sound.jl`, built on `testsound_viscous.jl`): eigenmode IC with genuinely 2D $\vec{k}$ ($n_x = n_y = 1$); ensemble-mean amplitude decay vs. $e^{-\Gamma_s k^2 t/2}$, $\Gamma_s = (\frac{4}{3}\eta/s + \zeta/s)/T_{\mathrm{rel},0}$; final profile vs. `analytic_M`; no fitting. *Parallel sweep* (optional): parallel particle loop, reduction into the accumulators; bit-identical to serial for the same key (App. B).